Inspiring Excellence

Program Outline

Inspiring Innovation

CVIS Hands-on Training on Development of Machine Learning and Deep Learning Models

SUMMER 2025

Target Audience:

Pre-thesis 2 undergraduate thesis students of the CVIS Lab aiming to enrich ML/DL-based model design and development skills for research works.

Training Information:

- **Starting Date:** 20 July 2025 (5.00 PM - 6.30 PM)
- **Duration:** 12 sessions: 1.5 hours each & 2 sessions per week (Saturday & Tuesday)
- **Location:** 12B22L
- **Mode:** Training Course (Hands-on lab + guided theory)
- **Tools:** Python, TensorFlow/Keras & PyTorch, Colab/Jupyter, Scikit-learn, HuggingFace, OpenCV, Weights & Biases/TensorBoard, Git

Contact:

Convenor: Dr. Md. Ashraful Alam, Associate Professor & Director of CVIS Research Lab
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Co-Convenor: Dr. Md. Golam Rabiul Alam, Professor & Director of CVIS Research Lab
CSE Department, BRAC University, Email: rabiul.alam@bracu.ac.bd

Coordinator: Sanjida Tasnim, Adjunct Lecturer & Faculty Member of CVIS Research Lab
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BRAC UNIVERSITY
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Sl.	Date	Main Subject	Topics / Content Covered	Facilitators
1	20-07-2025 Sunday	Introduction to ML/DL for Research	• Common ML/DL workflows • Overview of the model life cycle • Research model types & applications • Supervised vs Unsupervised • Semi-supervised learning • Course Overview	Dr. Md. Ashraful Alam Shammo Biswas Student Tutor (1)
2	22-07-2025 Tuesday	Data Science Foundations I	• Exploratory Data Analysis (EDA): distributions, correlations • Feature engineering • Handling missing values, outliers • Data scaling and normalization • Nearest neighbors • Bayesian classification • Regression	Dr. Md. Golam Rabiul Alam Anika Tahsin Student Tutor (1)
MIDTERM EXAM BREAK				
3	05-08-2025 Tuesday	Data Science Foundations II	• Support vector machines • Decision trees and forests • Clustering, K-means, and Gaussian mixture models • Principal component analysis	Dr. Md. Golam Rabiul Alam Anika Tahsin Student Tutor (1)
4	09-08-2025 Saturday	Deep neural networks	• Weights and layers • Activation functions • Regularization and normalization • Loss functions • Backpropagation • Training and optimization	Dr. Md. Ashraful Alam Shammo Biswas Student Tutor (1)
5	12-08-2025 Tuesday	Deep Learning for Computer Vision	• Data Pre-processing • Convolution • Pooling and unpooling • Application: Digit classification • Network architectures	Sanjida Tasnim Student Tutors (2)

			• Visualizing weights and activations • Adversarial examples	
6	16-08-2025 Saturday	Classification with CNNs	• Image preprocessing and augmentation • CNN architectures: VGG, ResNet, EfficientNet- Transfer learning and fine-tuning • Generative Adversarial Networks (GAN)	Sanjida Tasnim Student Tutors (2)
7	19-08-2025 Tuesday	Advanced Vision Tasks	• Object detection: YOLO, SSD • Segmentation: U-Net, DeepLab • Image annotation tools: CVAT, LabelImg- Visualizing predictions with Grad-CAM++	Shammo Biswas Student Tutors (2)
8	23-08-2025 Saturday	Deep Learning for Video Analysis	• Video preprocessing: OpenCV, FFmpeg • Frame extraction, temporal modeling • Models: 3D-CNNs, ConvLSTMs • Tasks: action recognition, captioning	Shammo Biswas Student Tutors (2)
9	26-08-2025 Tuesday	VLM Fundamentals & Transformers	• Text cleaning, tokenization • Word embeddings: Word2Vec, GloVe • Basic models: LSTM, GRU • Attention mechanisms • Intro to Transformer, BERT, GPT	Niloy Farhan Student Tutors (2)
10	30-08-2025 Saturday	Advanced VLM Tasks	• NER, text classification, summarization, QA • HuggingFace Transformers hands-on- Fine-tuning BERT • Metrics: BLEU, ROUGE, perplexity	Niloy Farhan Student Tutors (2)

11	02-09-2025 Tuesday	Multi-modal Learning	• Multi-modal data and challenges • Fusion strategies: early, late, hybrid • Architectures: CLIP, BLIP, LXMERT • Use cases: VQA, image captioning	Abdullah Sajjad Hasan Rafi Student Tutors (2)
12	06-09-2025 Saturday	Experiment Tracking & Explainability	• TensorBoard & Weights & Biases usage • Logging training metrics • Visualizing training/validation loss & confusion matrix • Explainability: Grad-CAM, SHAP, Saliency	Abdullah Sajjad Hasan Rafi Student Tutors (2)

Program Outline

List of Facilitators		
Name	**Designation**	**Role**
Dr. Md. Ashraful Alam (ASA)	Associate Professor & Director of CVIS Lab, CSE Department, BRAC University	• To conduct training sessions: Theoretical discussion & demonstration of ML/DL models • Evaluate and grade the home works
Dr. Md. Golam Rabiul Alam (GRA)	Professor & Director of CVIS Lab, CSE Department, BRAC University	• To conduct training sessions: Theoretical discussion & demonstration of ML/DL models • Evaluate and grade the home works
Sanjida Tasnim	Adjunct Lecturer, CSE Department, BRAC University	• To conduct training sessions: Theoretical discussion & demonstration of ML/DL models • Evaluate and grade the home works
Noloy Farhan	Lecturer, CSE Department, BRAC University	• To conduct training sessions: Theoretical discussion & demonstration of ML/DL models • Evaluate and grade the home works
Shammo Biswas	Research Assistant of CVIS Lab, CSE Department, BRAC University	• To conduct training sessions: Theoretical discussion & demonstration of ML/DL models • Evaluate and grade the home works
Anika Tahsin	Research Assistant of CVIS Lab, CSE Department, BRAC University	• To conduct training sessions: Theoretical discussion & demonstration of ML/DL models • Evaluate and grade the home works

Abdullah Sajjad Hasan Rafi	Research Assistant of CVIS Lab, CSE Department, BRAC University	● To conduct training sessions: Theoretical discussion & demonstration of ML/DL models ● Evaluate and grade the home works
Student Tutors (TBA)	Graduate/Senior Level Skilled Undergraduate Students, CSE Department, BRAC University	● To assist in training sessions: technical and software environment issues ● Consult training students outside of sessions on requests regarding training contents and programming resources